AptaRank: Interpretable Ranking and Target-Aware Annotation of Foundation-Model-Generated RNA Aptamers

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***Abstract*—General-purpose large language models or foundation models (FMs) trained on biological sequence data promise to accelerate the design of new biotechnologies, but new designs face an experimental validation bottleneck. RNA aptamer design is one such biotechnology application in which prioritising candidate aptamers is largely dependent on experimental data absent from FM settings. We present an interpretable, biologically grounded ranking and annotation tool intended to make computational prioritisation of FM-generated RNA aptamers more accessible. Using a two-tier ranking framework, we consciously separate intrinsic structural quality from target-aware geometric compatibility to provide complementary signals of candidate suitability. A researcher-facing dashboard presents key biological features alongside the ranked shortlist to inform decision-making. Our approach aims to support FM-assisted biotechnology development by helping to bridge the gap between AI-generated candidates, experimental validation, and production.**

***Index Terms*—RNA aptamers, foundation models, candidate ranking, interpretable machine learning, computational biology**

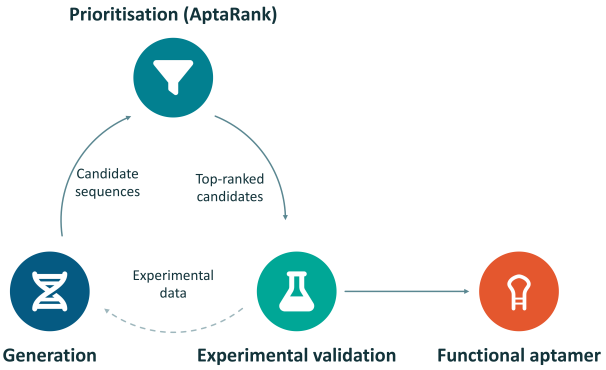


Fig. 1: Design–build–test–learn cycle for aptamer development: an RNA FM generates candidates, AptaRank prioritises them, and experimental results feed back into generation.

I. Introduction

RNA aptamers are small biological molecules that are useful as biosensors or therapeutics because of their ability to bind specific molecular targets [1]. These molecules are short, single-stranded sequences of RNA (ribonucleic acid) that fold into defined two- and three-dimensional structures, which bind to targets such as proteins with high specificity and affinity [1]. Historically, RNA aptamers have been designed using experimental approaches [1], but recent advances in the capabilities of large language models (LLMs) and foundation models (FMs) have offered a new strategy for their development [2]. Generative RNA FMs learn broad representations of RNA sequence data and have recently been applied to RNA aptamer design, producing thousands of plausible candidates [2]. However, for these FM-designed candidates to become functional aptamers, experimental validation is essential (Fig. 1). This validation is slow and costly, hence computational approaches are needed for filtering and prioritising candidates [1].

Various tools and frameworks have been developed for scoring, filtering, and ranking aptamer candidates. Sequence-based clustering and counting methods group candidates into families by sequence similarity metrics, then rank them by counts or experimental enrichment [3]. Structural motif tools incorporate predictions of two-dimensional structure, then rank candidates by combined sequence and structural stability scores [4]. Machine-learning- and deep-learning-based binding predictors use sequence representations of aptamers with their targets to predict an interaction probability [5]. Finally, structure-based docking and molecular dynamics simulations provide a detailed assessment of aptamer-target compatibility and binding stability [6]. The majority of these approaches

require experimental read counts, experimentally derived sequence pools, or labelled binding data, which do not exist for FM-generated candidates [1]. While docking avoids these caveats, it is computationally expensive and its reliability is sensitive to structural uncertainty [1]. These filtering frameworks remain useful for, but not tailored to, FM-designed aptamers.

AptaRank addresses these gaps by providing an FM-tailored, target-swappable, and biologist-interpretable prioritisation tool for RNA aptamers. It ranks candidate aptamers on their intrinsic structural quality in comparison to an aptamer reference distribution, then annotates the ranking by their control-relative geometric agreement to a protein target. This two-tier design separates target-agnostic quality from target-aware plausibility, accounting for the fundamental limits of target-aware scoring for RNA sequences [1]. Geometric agreement is binding-mode-aware, as aptamers bind to targets using more than one mechanism [1]. The tool explicitly reports plausibility, not binding prediction, computes shuffled-sequence control pass rates, and provides rule-based natural language explanations of outputs to enhance transparency.

In summary, this work introduces an interpretable, target-aware tool for filtering and ranking FM-generated RNA aptamers against user-specified targets. Our key contributions are: (1) A two-tier ranking framework for FM-generated RNA aptamer candidates; (2) A researcher-facing dashboard that highlights biologically significant features; and (3) A demonstration with FM-generated aptamers. By providing informative and biologically grounded metrics with which to prioritise FM-designed sequences, we hope to facilitate the use of RNA FMs in biotechnology applications.

II. System Architecture

AptaRank is designed as a two-tier framework integrating bioinformatics tools to rank and inform the prioritisation of candidate aptamer sequences by users. The architecture of the framework is illustrated in Fig. 2.

a) Tier 1: Tier 1 answers whether a sequence from the candidate data set is a well-formed, well-behaved, aptamer-like sequence. First, the ingested candidates are normalised, validated for admissible RNA alphabet and sequence length, deduplicated, and assigned stable IDs if not present. Then, the GC ratio is computed and scored with a two-sided percentile against the reference corpus, and ViennaRNA [7] is used to calculate normalised minimum free energy (MFE), ensemble defect, and the mean positional entropy. As RNA sequences can adopt multiple secondary structures, we sample many plausible structures (the number sampled depends on the user-specified run-rigour level), parse the elements that can protrude and contact a target using forgi, and report the distribution. Ensemble loop-size statistics are passed to Tier 2.

The composite Tier 1 score is a corpus-anchored, equally weighted mean of five per-criterion scores. Each raw feature x_i is mapped to a score $s_i \in [0,1]$ through the empirical cumulative distribution function F_i of that feature over the reference corpus. For normalised MFE, ensemble defect, and mean positional entropy, lower values are more favourable, so we use a one-sided sign-flipped percentile $s_i = 1 - F_i(x_i)$, while for stem fraction and GC ratio, where centrality within the reference distribution rather than magnitude is desired, we use a two-sided percentile $s_i = 1 - 2\,|F_i(x_i) - 0.5|$ that peaks at the corpus median. The composite is

$$\mathrm{Tier1} = \frac{\sum_i w_i s_i}{\sum_i w_i}, \qquad s_i \in [0,1],\ \mathrm{Tier1} \in [0,1], \quad (1)$$

with all weights $w_i = 1$ by default. F_i is the mid-rank empirical CDF over the reference corpus, so that tied values receive the average of their ranks.

Finally, a shuffled-control check assesses whether high structural scores actually reflect arrangement and not just composition. Each candidate is compared against dinucleotide-preserving shuffles of itself, for which the ViennaRNA [7] and forgi calculations are repeated and the corpus-calibrated structural sub-score (the mean of normalised MFE, ensemble defect, mean positional entropy, and stem fraction) is recomputed. GC ratio and length are excluded, as dinucleotide shuffling preserves them exactly. Significance is a Monte Carlo p-value with a pseudocount, $p = (1+c)/(M+1)$, where c is the number of the M shuffles scoring at least as high as the candidate; a candidate passes when $p \leq 0.05$.

b) Tier 2: Tier 2 provides a signal of aptamer-target compatibility by asking, given a target and binding mode, whether each candidate's shape could plausibly fit the target binding site. According to the user-specified target configuration, AptaRank first fetches the experimental or predicted structure from the Protein Data Bank (PDB) or AlphaFoldDB, selects the target chain and removes any others, and strips water and, optionally, other hetero-groups without changing the conformation of the selected chain. A single fpocket [8] run characterises candidate binding sites on the target by their volume, characteristic dimension, equivalent-sphere diameter, elongation, and planarity. When binding-site residues are input, the site is selected by overlap with them, otherwise the highest-scoring pocket is chosen automatically.

In pocket-binding mode, the aptamer presents a flexible loop or unpaired region that engages a concave pocket on the target. We compute the pocket's characteristic dimension d_{pocket} (Å) as the largest principal-axis extent of the fpocket alpha-sphere cloud, and the aptamer's estimated accessible loop extent $d_{\mathrm{apt}} = a\,\bar{\ell}$ (Å), where $\bar{\ell}$ is the median ensemble loop size and $a = 6.0$ Å/nt. The agreement score is $\exp\!\left(-(d_{\mathrm{apt}} - d_{\mathrm{pocket}})^2 / 2\sigma^2\right)$ with bandwidth $\sigma = 6.0$ Å. The equivalent-sphere diameter is retained as a cross-check on pocket shape.

In surface-binding mode, the aptamer covers a relatively planar surface patch via shape and charge complementarity. AptaRank requires user-specified binding-site residues and reuses the single fpocket [8] run, defining the patch as those residues and its area as their total solvent-accessible surface area computed with freeSASA on the isolated target chain, with planarity given by the smallest principal-axis extent of the patch-atom coordinates. The size-coverage term is

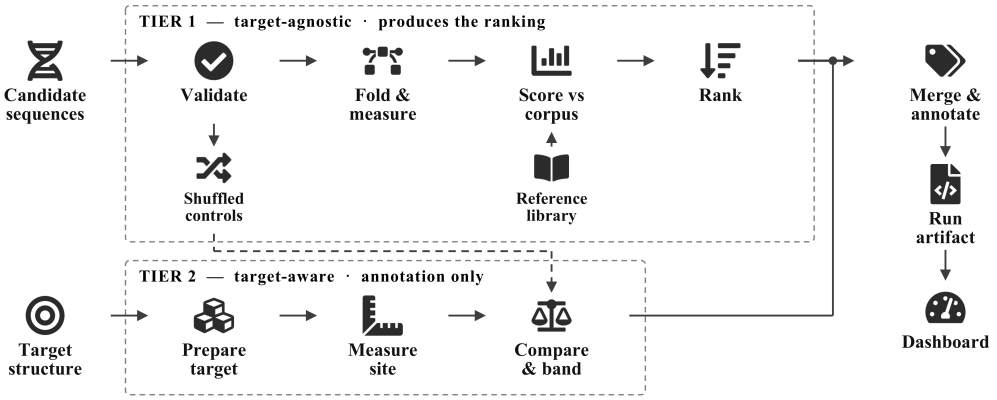


Fig. 2: AptaRank system architecture.

calculated as a Gaussian agreement $\exp\left(-\Delta^2/2\sigma^2\right)$ on the area mismatch $\Delta = |A_{\text{footprint}} - A_{\text{patch}}|$, with bandwidth $\sigma = 600$ Å^2. RNA footprint area $A_{\text{footprint}}$ is estimated from the median radius of gyration over the sampled structural ensemble, derived from the forgi element graph. The reported Tier 2 score is the percentile of the candidate's geometric mismatch against a fixed bank constructed by generating an equal number of deterministically seeded, dinucleotide-preserving shuffles from every unique sequence in the reference corpus and computing their structural descriptors once for reuse across targets. Mean electrostatic potential over the patch is computed and reported separately as a target-level signal, with per-candidate charge integration planned for future work.

c) Interpretable dashboard: The Tier 2 control percentile is banded into weak ($<$ 75th), moderate (75th–95th), and strong ($>$ 95th) geometric compatibility. All candidates are ranked by composite Tier 1 score, then the top 1000 candidates are annotated with their banded Tier 2 scores and shuffled-control pass/fail indicators. This explicit separation of ranking from annotation prevents mediocre candidates with flattering target-plausibility scores from outranking structurally excellent aptamers.

III. Demonstration and Evaluation

A. Demonstration steps

Step 1: User uploads aptamer data set(s) for analysis and ranking. The user provides a file (.txt or .csv) containing candidate RNA sequences of length 20–100 nucleotides and characters A, C, G, U (T also accepted) for comparison of intrinsic structural quality against a reference data set. Candidates are compared against a reference library of validated aptamers with associated target provenance. Users may also upload a study-specific reference library.

Step 2: Optional selection of protein target and specification of binding mode and site. Users input either the PDB ID or UniProt ID corresponding to the experimentally determined or AlphaFold-predicted structure of their protein target respectively, specify the exact chain of interest, and select a binding mode. They optionally also define the binding site amino acid residues (required for surface-binding).

Step 3: Choose run rigour level and start analysis. Users choose between a quick preview run (no shuffled controls performed, only 30 sampled structures per sequence, no shuffled-control outcomes), a standard run (20 shuffled controls and 100 sampled structures), and a rigorous run (99 shuffled controls and 200 sampled structures).

Step 4: Results are displayed on the interactive dashboard. The dashboard, which presents the ranked and annotated candidate list, is depicted in Fig. 3. Upon user selection, individual candidates display their Tier 1 score breakdown across categories, composite Tier 1 score, Tier 2 and shuffled control status, length, sequence, dot-bracket structure, ViennaRNA-predicted secondary structure, and natural language explanation of the score and rank. When a target is configured, the dashboard also presents a Tier 1–Tier 2 scatter plot, a batch-level Spearman correlation as descriptive diagnostics, and a target panel identifying the specified protein, binding site area measurements, binding-mode, site residues, and electrostatics for surface binding mode.

B. Preliminary evaluation

We assessed whether Tier 1 assigns higher scores to experimentally supported aptamers held out from its reference set than to length-matched random RNA. Protein-target aptamers curated from seven public sources were exact-core deduplicated, clustered using MMseqs2 at 90% identity and 80% coverage, and assigned by intact clusters while balancing target, evidence, length, and source. This produced 113 reference and 95 held-out unique cores with no core or cluster overlap. Each held-out core was paired with a uniformly sampled A/C/G/U control of identical length. Held-out aptamers scored higher than the paired controls (median 0.531 [IQR 0.420–0.645] versus 0.430 [0.295–0.550]; Wilcoxon $W = 1304$, $p = 0.00029$; rank-biserial effect $= 0.428$). However, shuffle-pass rates were low and did not favour the aptamers (5/95 versus 7/95), supporting a modest target-agnostic signal with limited sequence-level discrimination, not evidence of binding.

Because score-blind screening identified only one eligible target system, Tier 2 was evaluated illustratively using two GRK2 cognates, four length-matched noncognates, and two

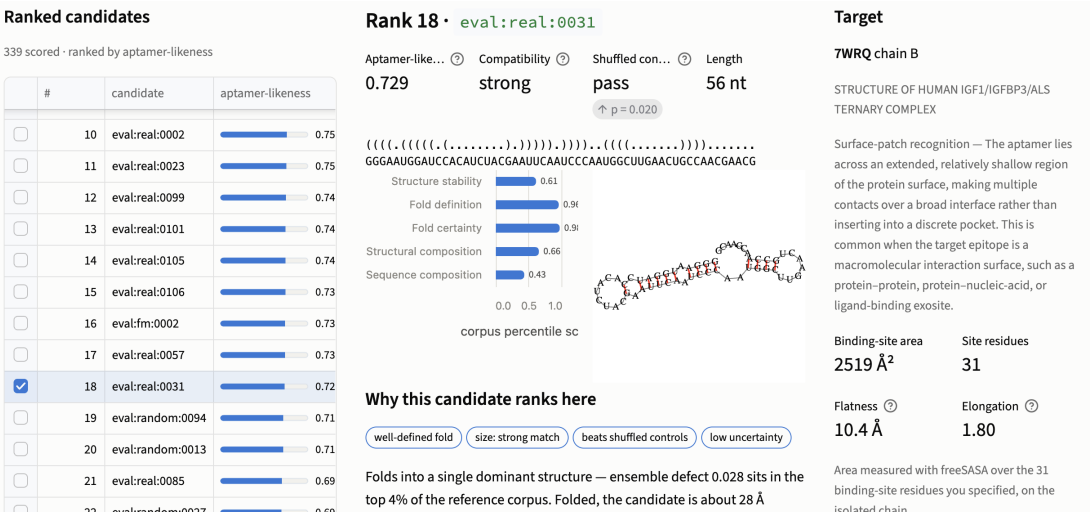


Fig. 3: AptaRank dashboard user interface.

uniform controls against the GRK2 structures 3UZS and 3UZT in pocket-binding mode. Both cognates were weak under both structures, no strong band occurred, and three of eight candidates changed band. This null/mixed result shows sensitivity to the selected bound-state structure and does not validate binding or non-binding.

Runs used AptaRank 0.1.0, ViennaRNA 2.7.2 [7], and forgi 2.2.3, with 200 sampled structures, 99 per-candidate Tier 1 shuffles, and seed 42; Tier 2 additionally used fpocket 4.0. Limitations include a similarity-aware but not fully family-aware split and a single-target Tier 2 evaluation without independent binding evidence.

TABLE I: Tier 1 aptamer-likeness across the two test groups.

Group	n	Tier 1 median [IQR]	Shuffle pass
Held-out aptamers	95	0.531 [0.420–0.645]	5/95 (5.3%)
Exact-length controls	95	0.430 [0.295–0.550]	7/95 (7.4%)

IV. Conclusion and Future Work

This paper has introduced a tool for interpretable ranking and prioritisation of FM-generated candidate RNA aptamers against user-specified protein targets. AptaRank's two-tier approach performs both target-agnostic evaluation of intrinsic structural quality and target-swappable annotation of geometric compatibility. The source code of the framework, along with a demonstration video, is publicly available[1]. Future work will implement additional binding modes and extend evaluation to be family-aware, consider multi-target panels, and independently measure binding outcomes. Our work hopes not only to facilitate more accessible computational prioritisation of aptamers, but also to encourage the integration of FMs into design–build–test–learn cycles for biotechnology development.

[1]https://github.com/alishahsvnd/AptaRank

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